

Linear Equations with constant coefficients

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26. $(D^2 + D + 1)y = e^{3x} + 6e^x - 3e^{-2x} + 5$

$PI = \frac{1}{D^2 + D + 1} \times (e^{3x} + 6e^x - 3e^{-2x} + 5e^{0x})$

$= \frac{1}{D^2 + D + 1} \times e^{3x} + \frac{1}{D^2 + D + 1} \times 6e^x - \frac{1}{D^2 + D + 1} \times 3e^{-2x} + \frac{1}{D^2 + D + 1} \times 5e^{0x}$

$= \frac{e^{3x}}{13} + 2e^x - e^{-2x} + 5$

$CF = e^{-\frac{1}{2} \pm \frac{\sqrt{3}}{2}i}$
 $CF = e^{-\frac{x}{2}} (c_1 \cos \frac{\sqrt{3}}{2}x + c_2 \sin \frac{\sqrt{3}}{2}x)$
 $y = CF + PI$
 $= e^{-\frac{x}{2}} (c_1 \cos \frac{\sqrt{3}}{2}x + c_2 \sin \frac{\sqrt{3}}{2}x)$
 $+ \frac{e^{3x}}{13} + 2e^x - e^{-2x} + 5$
Ans.

27. $D^2y - y = e^x$

$PI = \frac{1}{D^2 - 1} \times e^x$

$= \frac{x}{2D} \times e^x$

$= \frac{xe^x}{2}$

$D = 1, -1$

$CF = c_1 e^x + c_2 e^{-x}$

$y = PI + CF$

$= c_1 e^x + c_2 e^{-x} + \frac{xe^x}{2}$
Ans.

28. $(D-2)^2 y = e^x + xe^{2x}$

$PI = \frac{1}{(D-2)^2} \times e^x + \frac{1}{(D-2)^2} \times xe^{2x}$

$= e^x + e^{2x} \cdot \frac{1}{(D-2)^2} \times x$

$= e^x + e^{2x} \cdot \frac{1}{(D+2-2)^2} \times x$

$= e^x + e^{2x} \cdot \frac{1}{D^2} \times x$

$= e^x + e^{2x} \cdot \frac{x^3}{6}$

~~CF~~ $D = 2, 2$

$CF = (c_1 x + c_2) e^{2x}$

$y = CF + PI = (c_1 x + c_2) e^{2x} + e^x + e^{2x} \cdot \frac{x^3}{6}$
Ans.

29. $(D^4 - 1)y = \sin 2x$

$PI = \frac{1}{D^4 - 1} x \sin 2x$

$= \frac{1}{(D^2)^2 - 1} x \sin 2x$

$= \frac{1}{(-4)^2 - 1} x \sin 2x$

$= \frac{\sin 2x}{15}$

$y = CF + PI = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x + \frac{\sin 2x}{15}$ Ans.

30. $(D^3 + 1)y = \cos x$

$PI = \frac{1}{D^3 + 1} x \cos x$

$= \frac{1}{D \cdot D^2 + 1} x \cos x$

$= \frac{1}{-D + 1} x \cos x$

$= \frac{(1+D)}{(1-D)(1+D)} x \cos x$

$= \frac{1+D}{1^2 - D^2} \cos x$

$= \frac{1+D}{2} \cos x$

$= \frac{1}{2} + \frac{1}{2} (-\sin x)$

$= \frac{1}{2} \cos x + \frac{1}{2} (-\sin x)$

$= \frac{1}{2} (\cos x - \sin x)$

$D = -1, \frac{1}{2} \pm \frac{\sqrt{3}}{2} i$

$CF = c_1 e^{-x} + e^{\frac{x}{2}} \left[c_2 \cos \frac{\sqrt{3}}{2} x + c_3 \sin \frac{\sqrt{3}}{2} x \right]$

$y = CF + PI$
 $= c_1 e^{-x} + e^{\frac{x}{2}} \left(c_2 \cos \frac{\sqrt{3}}{2} x + c_3 \sin \frac{\sqrt{3}}{2} x \right) + \frac{1}{2} (\cos x - \sin x)$ Ans.

31. $(D^2 + 4)y = \sin 2x$

$D = \pm 2i$

$PI = \frac{1}{D^2 + 4} x \sin 2x$

$CF = c_1 \cos 2x + c_2 \sin 2x$

$= \frac{x}{2D} \sin 2x$

$= \frac{x}{2} \cdot \frac{1}{D} \sin 2x$

$= \frac{x}{2} \int \sin 2x$

$= \frac{x}{2} \cdot \frac{-\cos 2x}{2}$

$= -\frac{x}{4} \cos 2x$

$y = CF + PI = c_1 \cos 2x + c_2 \sin 2x + \frac{x(-\cos 2x)}{4}$ Ans.

32. $(D^2 + 5)y = \cos \sqrt{5} x$

$PI = \frac{1}{D^2 + 5} x \cos \sqrt{5} x$

$= \frac{x}{2D} x \cos \sqrt{5} x$

$= \frac{x}{2D^2} x D \cos \sqrt{5} x$

$= \frac{x}{-10} (-\sin \sqrt{5} x) \cdot \sqrt{5}$

$= \frac{\sqrt{5} x \sin \sqrt{5} x}{10}$

$D = \pm \sqrt{5} i$

$CF = c_1 \cos \sqrt{5} x + c_2 \sin \sqrt{5} x$

$y = CF + PI = c_1 \cos \sqrt{5} x + c_2 \sin \sqrt{5} x + \frac{\sqrt{5}}{10} x \sin \sqrt{5} x$ Ans.

$$33. (D^3 + D^2 + D + 1)y = e^x + e^{-x} + \sin x$$

$$D = 1, -1$$

$$CF = C_1 e^x + C_2 e^{-x}$$

$$(PI)_1 = \frac{1}{D^3 + D^2 + D + 1} x e^x$$

$$= \frac{e^x}{4}$$

$$(PI)_2 = \frac{1}{D^3 + D^2 + D + 1} x e^{-x}$$

$$= \frac{x}{3D^2 + 2D + 1} x e^{-x}$$

$$= \frac{x e^{-x}}{2}$$

$$(PI)_3 = \frac{1}{D^3 + D^2 + D + 1} x \sin x$$

$$= \frac{1}{D^2 + D^2 + D + 1} \sin x$$

$$= \frac{x}{3D^2 + 2D + 1} x \sin x$$

$$= \frac{x}{2D - 2} x \sin x$$

$$= \frac{x}{2(D-1)} x \sin x$$

$$= \frac{x}{2} \cdot \frac{(D+1)}{D^2 - 1} \sin x$$

$$= -\frac{x}{4} (\sin x + D \sin x)$$

$$= -\frac{x}{4} (\sin x + \cos x)$$

$$PI = (PI)_1 + (PI)_2 + (PI)_3$$

$$= \frac{e^x}{4} + \frac{x e^{-x}}{2} - \frac{x}{4} (\sin x + \cos x)$$

$$D = -1, \pm i$$

$$CF = C_1 e^{-x} + C_2 \cos x + C_3 \sin x$$

$$y = CF + PI = C_1 e^{-x} + C_2 \cos x + C_3 \sin x + \frac{e^x}{4}$$

$$+ \frac{x e^{-x}}{2} - \frac{x}{4} (\sin x + \cos x)$$

$$34. (D^2 - 1)y = x^2$$

$$PI = \frac{1}{D^2 - 1} x x^2$$

$$= -\frac{1}{1 - D^2} x x^2$$

$$= -\frac{1}{(1+D)(1-D)} x x^2$$

$$= - (1+D)^{-1} (1-D)^{-1} x^2$$

$$= - [(1-D+D^2)(1+D+D^2)] x^2$$

$$= - [1+D+D^2-D-D^2+D^2] x^2$$

$$= - [x^2 + D^2 x^2]$$

$$= - [x^2 + 2]$$

$$= -x^2 - 2$$

$$y = CF + PI = C_1 e^x + C_2 e^{-x} - x^2 - 2$$

$$35. D^4(D^2 - 1)y = x^2$$

Ans.

$$\Rightarrow (D^6 - D^4)y = x^2$$

$$D = 0, 0, 0, 0, \pm 1$$

$$CF = C_1 + C_2 x + C_3 x^2 + C_4 x^3 + C_5 e^x + C_6 e^{-x}$$

$$PI = \frac{1}{D^6 - D^4} x x^2$$

$$= -\frac{1}{D^4(1-D^2)} x x^2$$

$$= -\frac{1}{D^4} (1-D^2)^{-1} x x^2$$

$$= -\frac{1}{D^4} (1+D)^{-1} (1-D)^{-1} x x^2$$

$$= -\frac{1}{D^4} [(1-D+D^2)(1+D+D^2)] x^2$$

$$= -\frac{1}{D^4} (1+D^2) x^2$$

$$= -\frac{1}{D^4} (x^2 + 2)$$

$$= -\frac{x^6}{360} - \frac{1}{12} x^4$$

Ans.

$$D = \pm \sqrt{2}i$$

$$CF = c_1 \cos \sqrt{2}x + c_2 \sin \sqrt{2}x$$

36. $(D^2+2)y = x^3 + x^2 + e^{-2x} + \cos 3x$

$$(PI)_1 = \frac{1}{D^2+2} x x^3$$

$$= \frac{1}{2[1+\frac{D^2}{2}]} x x^3$$

$$= \frac{1}{2} x \left[1 + \frac{D^2}{2}\right]^{-1} x x^3$$

$$= \frac{1}{2} \left[1 - \frac{D^2}{2} + \left(\frac{D^2}{2}\right)^2\right] x^3$$

$$= \frac{1}{2} \left[x^3 - \frac{D^2}{2} x^3\right]$$

$$= \frac{1}{2} \left[x^3 - \frac{6x}{2}\right]$$

$$= \frac{1}{2} [x^3 - 3x]$$

$$(PI)_2 = \frac{1}{D^2+2} x x^2$$

$$= \frac{1}{2[1+\frac{D^2}{2}]} x x^2$$

$$= \frac{1}{2} \left[1 + \frac{D^2}{2}\right]^{-1} x x^2$$

$$= \frac{1}{2} \left[1 - \frac{D^2}{2} + \left(\frac{D^2}{2}\right)^2\right] x^2$$

$$= \frac{1}{2} \left[x^2 - \frac{1}{2} \cdot 2\right]$$

$$= \frac{1}{2} (x^2 - 1)$$

$$(PI)_3 = \frac{1}{D^2+2} x e^{-2x}$$

$$= \frac{e^{-2x}}{6}$$

$$(PI)_4 = \frac{1}{D^2+2} x \cos 3x$$

$$= -\frac{\cos 3x}{7}$$

$$PI = \frac{1}{2} (x^3 - 3x) + \frac{1}{2} (x^2 - 1) + \frac{e^{-2x}}{6} - \frac{\cos 3x}{7}$$

$$= \frac{1}{2} (x^3 - 3x + x^2 - 1) + \frac{1}{6} e^{-2x} - \frac{1}{7} \cos 3x$$

$$\therefore y = CF + PI$$

Ans.

37. $(D^2 - 2D - 1)y = e^x \cos x$ $|_{D = -1 \pm \sqrt{2}}$

$$PI = \frac{1}{D^2 - 2D - 1} e^x \cos x$$

$$= e^x \frac{1}{(D+1)^2 - 2(D+1) - 1} \cos x$$

$$= e^x \frac{1}{D^2 + 2D + 1 - 2D - 2 - 1} \cos x$$

$$= e^x \frac{1}{D^2 - 2} \cos x$$

$$= \frac{e^x \cos x}{-3}$$

$$CF = c_1 e^{(1+\sqrt{2})x} + c_2 e^{(1-\sqrt{2})x}$$

38. $(D-2)^2 y = e^{2x} \cdot x^{-2}$ $y = CF + PI$ Ans.

$$PI = \frac{1}{(D-2)^2} e^{2x} \cdot x^{-2}$$

$$= e^{2x} \frac{1}{(D+2-2)^2} x^{-2}$$

$$= e^{2x} \frac{1}{D^2} x^{-2}$$

$$= -\log x \cdot e^{2x}$$

$$D = 2, 2$$

$$CF = (c_1 x + c_2) e^{2x}$$

$$y = (c_1 x + c_2) e^{2x} - \log x \cdot e^{2x}$$

Ans.

39. $(D^2 - 1)y = xe^{3x}$

$$PI = \frac{1}{D^2 - 1} \times x \cdot e^{3x}$$

$$= e^{3x} \frac{1}{(D+3)^2 - 1} \cdot x$$

$$= e^{3x} \frac{1}{D^2 + 6D + 9 - 1} \cdot x$$

$$= e^{3x} \frac{1}{D^2 + 6D + 8} x$$

$$= e^{3x} \frac{1}{8 \left(\frac{D^2}{8} + \frac{3}{4}D + 1 \right)} x$$

$$= e^{3x} \frac{1}{8} \cdot \left(1 + \frac{D^2 + 6D}{8} \right)^{-1} \cdot x$$

$$= \frac{e^{3x}}{8} \left[1 - \left(\frac{D^2 + 6D}{8} \right) + \left(\frac{D^2 + 6D}{8} \right)^2 \right] x$$

$$= \frac{e^{3x}}{8} \left[x - \frac{3}{4} Dx \right]$$

$$= \frac{e^{3x}}{8} \left(x - \frac{3}{4} \right)$$

$$= \frac{1}{32} e^{3x} (4x - 3)$$

$$D = 1, -1$$

$$CF = C_1 e^x + C_2 e^{-x}$$

$$y = CF + PI = \frac{1}{32} e^{3x} (4x - 3) + C_1 e^x + C_2 e^{-x}$$

Ans.

40. $(D^2 + 5D + 6)y = e^{-2x} (\sec^2 x) (1 + 2 \tan x)$

$$D^2 + 5D + 6 = 0$$

$$D = -2, -3$$

$$CF = C_1 e^{-2x} + C_2 e^{-3x}$$

Here,

$$y_1 = e^{-2x}$$

$$y_2 = e^{-3x}$$

$$X = e^{-2x} (\sec^2 x) (1 + 2 \tan x)$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

$$= \begin{vmatrix} e^{-2x} & e^{-3x} \\ -2e^{-2x} & -3e^{-3x} \end{vmatrix}$$

$$= -3e^{-3x} \cdot e^{-2x} + 2e^{-3x} \cdot e^{-2x}$$

$$= -e^{-5x}$$

$$\begin{aligned}
 u_1 &= - \int \frac{y_2 X}{W} dx \\
 &= - \int \frac{e^{-3x} e^{-2x} \sec^2 x (1+2 \tan x)}{-e^{-5x}} dx \\
 &= \int \sec^2 x (1+2 \tan x) dx \\
 &= \int (1+2u) du \\
 &= u + u^2 \\
 &= \tan x + \tan^2 x
 \end{aligned}$$

$$\tan x = u$$

$$\frac{du}{dx} = \sec^2 x$$

$$\Rightarrow du = \sec^2 x dx$$

$$D = -2, -3$$

$$CF = c_1 e^{-2x} + c_2 e^{-3x}$$

$$\begin{aligned}
 u_2 &= \int \frac{y_1 X}{W} dx \\
 &= \int \frac{e^{-2x} e^{-2x} \sec^2 x (1+2 \tan x)}{-e^{-5x}} dx \\
 &= - \int e^x \sec^2 x (1+2 \tan x) dx
 \end{aligned}$$

$$Ans, u_2 = -e^x \sec^2 x = -(e^x \sec^2 x)$$

$$PI = u_1 y_1 + u_2 y_2 = e^{-2x} (\tan x + \tan^2 x) - e^{-3x} (e^x \sec^2 x) = e^{-2x} + e^{-2x} \tan x$$

~~As, $u_2 y_2$ is not linearly independent from y_2 ,
 so, $u_2 y_2 = 0$ Here, $-e^{-2x}$ already belongs to CF.~~

$$\text{Thus, } PI = e^{-2x} \tan x$$

$$y = PI + CF = e^{-2x} \tan x + c_1 e^{-2x} + c_2 e^{-3x} \quad \underline{\text{Ans.}}$$

Alternate Method:

$$\underline{Q0.} (D^2 + 5D + 6) y = e^{-2x} (\sec^2 x) (1+2 \tan x)$$

$$PI = e^{-2x} \frac{1}{(D-2)^2 + 5(D-2) + 6} \cdot \sec^2 x (1+2 \tan x)$$

$$= e^{-2x} \frac{1}{D^2 - 4D + 4 + 5D - 10 + 6} \cdot \sec^2 x (1+2 \tan x)$$

$$= e^{-2x} \frac{1}{D^2 + D} \sec^2 x (1+2 \tan x)$$

$$= e^{-2x} \frac{1}{D(D+1)} \sec^2 x (1+2\tan x)$$

$$= e^{-2x} \frac{1}{D+1} \int \sec^2 x (1+2\tan x) dx$$

$$= e^{-2x} \frac{1}{D+1} \int (1+2u) du$$

$$= e^{-2x} \frac{1}{D+1} \cdot u + u^2$$

$$= e^{-2x} \frac{1}{D+1} \tan x + \tan^2 x$$

$$= e^{-2x} \left[e^{-x} \int e^x (\tan x + \tan^2 x) dx \right]$$

$$= e^{-2x} \left[e^{-x} \cdot e^x (\tan x + \tan^2 x) \right]$$

$$= e^{-2x} (\tan x + \tan^2 x)$$

$$= e^{-2x} \tan x + e^{-2x} \tan^2 x$$

$$= e^{-2x} \tan x$$

$$\begin{aligned} \tan x &= u \\ \frac{du}{dx} &= \sec^2 x \\ du &= \sec^2 x dx \end{aligned}$$

$$D = -2, -3$$

$$CF = c_1 e^{-2x} + c_2 e^{-3x}$$

$$\therefore y = CF + PI$$

$$= c_1 e^{-2x} + c_2 e^{-3x} + e^{-2x} \tan x$$

Ans.